DATA ANALYSIS AND PREDICTIVE MODELING

DIBS RETAIL COMPANY

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INTRODUCTION

Dibs, a rapidly growing online retail company, seeks to overcome challenges in increasing sales and fostering customer loyalty. To achieve this, the company has engaged our team to analyze sales data and provide actionable insights for refining sales and marketing strategies.

Utilizing advanced data visualization and statistical techniques, our team aims to uncover patterns and trends in customer behavior. By delving into sales history data, we seek to identify customer preferences, buying patterns, and potential areas for improvement in product offerings and marketing approaches.

Furthermore, harnessing the power of machine learning methodologies, our primary objective is to develop a predictive model capable of forecasting sales. By projecting sales quantity and incorporating product price, we can generate forecasted sales figures. This predictive capability will empower Dibs to proactively tailor sales and marketing efforts, anticipate customer needs, personalize interactions, and elevate overall customer satisfaction and loyalty.

Ultimately, our goal is to deliver actionable recommendations that will enable Dibs to optimize its sales and marketing strategies, drive revenue growth, and cultivate enduring customer relationships in the competitive online retail landscape.

METHODOLOGY

This project adopted a structured analytical approach to examine DIBS Retail Company's sales data, uncover performance patterns, and develop a predictive model to support strategic decision-making.

Data Overview

The dataset comprises monthly sales records from January to December, covering six fields:

- Order ID - A unique identifier to keep track of orders.
- Product - the item(s) sold to the customer.
- Quantity Ordered - The total quantity of items ordered, prior to any modifications.
- Price Each - The price of each individual product.
- Order Date - The date on which the customer requests their order to be shipped.
- Purchase Address - The address obtained from the purchase order.

Data Preparation and Cleaning

A comprehensive cleaning process was applied to ensure data integrity

- **Combining Datasets**: Sales data from 12 separate files, one for each month of the year, were combined into a single dataset for comprehensive analysis
- **Data cleaning and transformation**: To ensure data quality and readiness for analysis, several data cleaning steps were performed
    - Type Conversion:
        - i. Convert 'Order ID', 'Quantity Ordered', and 'Price Each' into numeric formats.
        - ii. Convert 'Order Date' into date format.
    - Data Transformation:
        - i. Decompose the 'Order Date' into separate fields: date, month, year, hour, and a binary indicator for weekends, and holidays.
        - ii. Break down the 'Purchase Address' into city, state, and postcode.
    - Outliers Handling: Adjust outlier values to mitigate their impact on the analysis.
    - Calculation: Compute the total sales for each order by multiplying the 'Quantity Ordered' by the 'Price Each'.

Exploratory Data Analysis (EDA)

EDA was conducted to understand the shape, distribution, and quality of the consolidated dataset. Summary statistics and frequency distributions were examined across products, cities, and time periods to identify early patterns ahead of deeper analysis.

Deep Analysis and Visualization

Deep analysis was performed across three dimensions: time, geography, and product.

- Time analysis revealed strong Q4 seasonality, with December peaking at 28,137 orders and $4.6M in sales, and identified two daily peak ordering windows (11AM–1PM and 6PM–8PM).
- Geographic analysis showed California driving 39.8% of total sales, with San Francisco as the top-performing city. Product analysis found that accessories dominated order quantities, while premium electronics led revenue contribution at $8M (22% of total sales).
- Frequently co-purchased product pairs — notably iPhone and Lightning Charging Cable — were also identified.

Predictive Modelling

K-means clustering was first applied to segment products by quantity and unit price, yielding two clusters: Cluster 1 - high quantity, low price and Cluster 2 - low quantity, high price

These cluster labels were then incorporated as features into two predictive models: Linear Regression and Decision Tree, both trained to forecast sales quantity using price and cluster as predictors. Model performance was evaluated using MAE, MSE, and R-squared.

The Decision Tree outperformed Linear Regression (MAE: 0.046 vs 0.048; R^2: 0.776 vs 0.764) and was selected as the preferred forecasting model. Combining predicted quantity with product price enables generation of projected sales figures.

Insights and Recommendations

Recommendations were developed across four strategic areas based on analytical findings.

- **Inventory management**: demand forecasting using the Decision Tree model was proposed to ensure adequate stock of high-turnover items (e.g., AAA Batteries) during peak periods, while low-performing SKUs such as LG Dryers are flagged for reduction or discontinuation.
- **Sales strategy**: seasonal and time-based promotions are recommended during Q4 and peak ordering hours, along with bundle deals leveraging frequently co-purchased product pairs.
- **Marketing**: targeted campaigns should focus on high-value regions (San Francisco, California) and peak ad windows, with cross-selling strategies based on purchase behaviour.
- **Market expansion**: entering densely populated, geographically proximate states including Florida, Pennsylvania, and Illinois is recommended to extend market reach while leveraging existing warehouse and logistics infrastructure.

EXPLORATORY DATA ANALYSIS

Key highlights of this report:

1. Sales Performance:

- Worst Year: 2021.
- Best Year: 2019.
- Best month: December.
- Top Sales City: San Francisco.
- Optimal Advertising Time: 19:00.

2. Product Insights:

- Most Sold Product: AAA Batteries (4-pack).

Top 5 Products	Quantity	Price/unit
AAA Batteries	31,017	2
AA Batteries	27,635	3
USB-C Charging cable	23,974	11
Lightning charging cable	20,557	14
Wired headphones	15,661	11

Top selling products are mostly low-priced items with values ranging from $2 to $11.

- Least Sold Product: LG Dryer.

Least Selling Products	Quantity	Price/unit
LG Dryer	646	600
LG Washing Machine	666	600
Vareebadd Phone	2,068	400
20in Monitor	4,129	109
ThinkPad laptop	4,132	999

- Products Sold Together: iPhone and lightning charging cable.

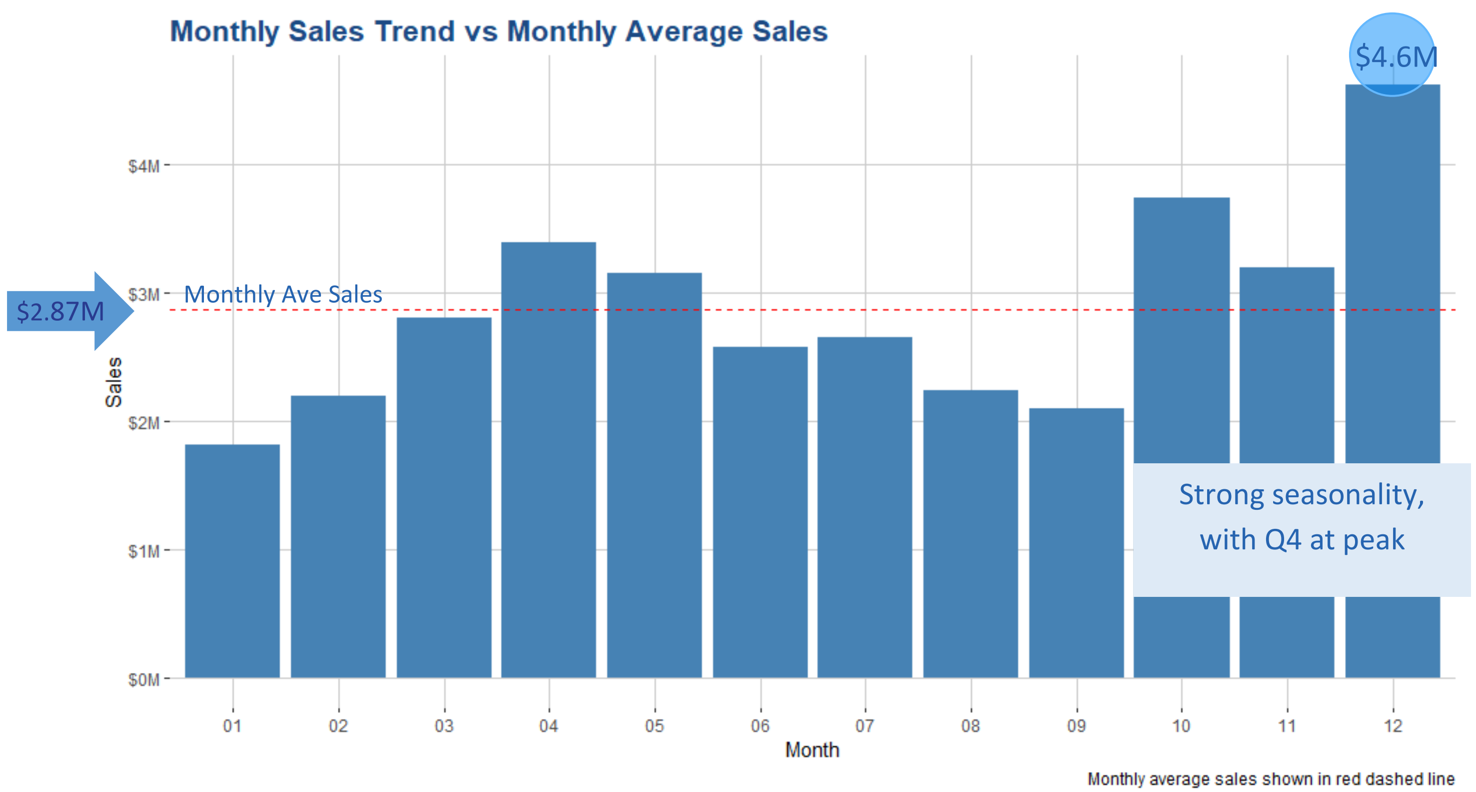
Combination	Quantity
iPhone and lightning charging cable	1002
Google phone and USB-C charging cable	985
iPhone and wired headphones	447
Google phone and wired headphones	413
Vareebadd phone and USB-C charging cable	361

DEEP ANALYSIS AND VISUALIZATION

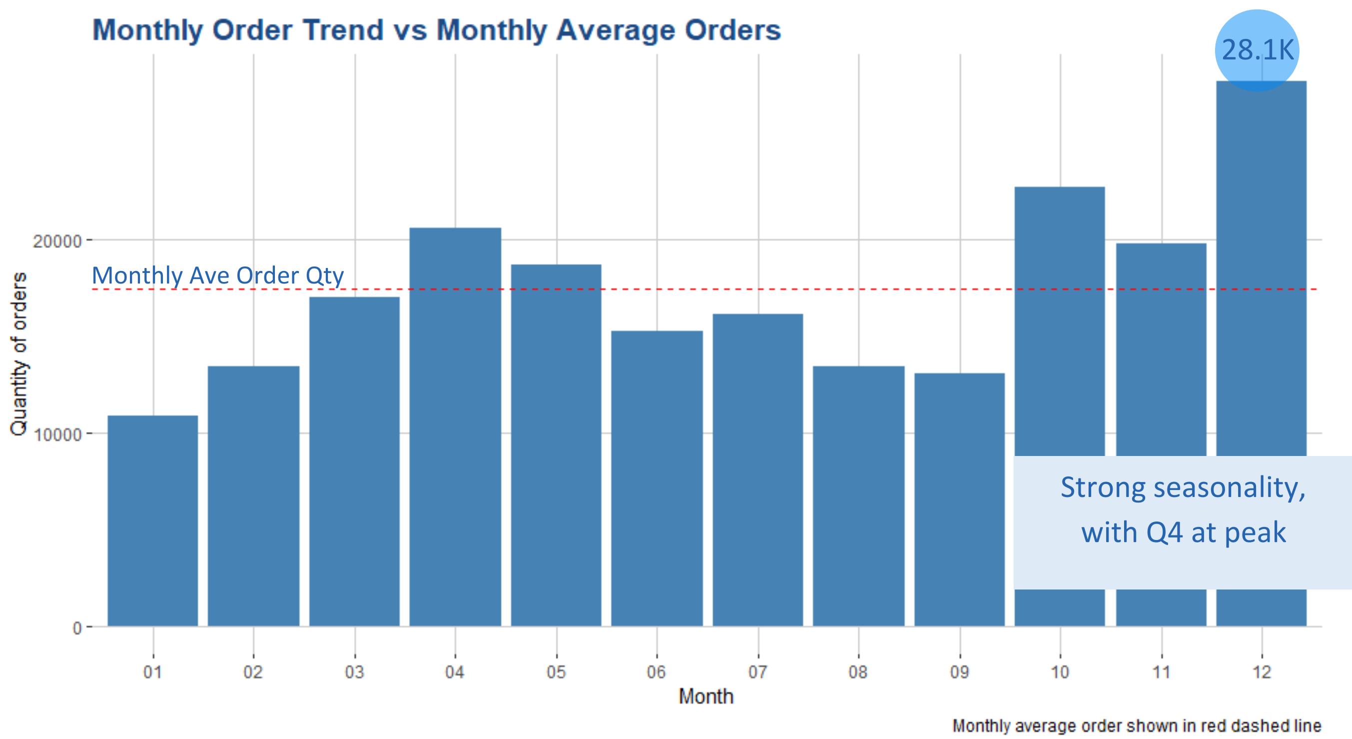
In this section, we delve into Dibs' business performance during the year 2019, leveraging visualizations to gain insights. Let's explore key aspects:

Timely basic

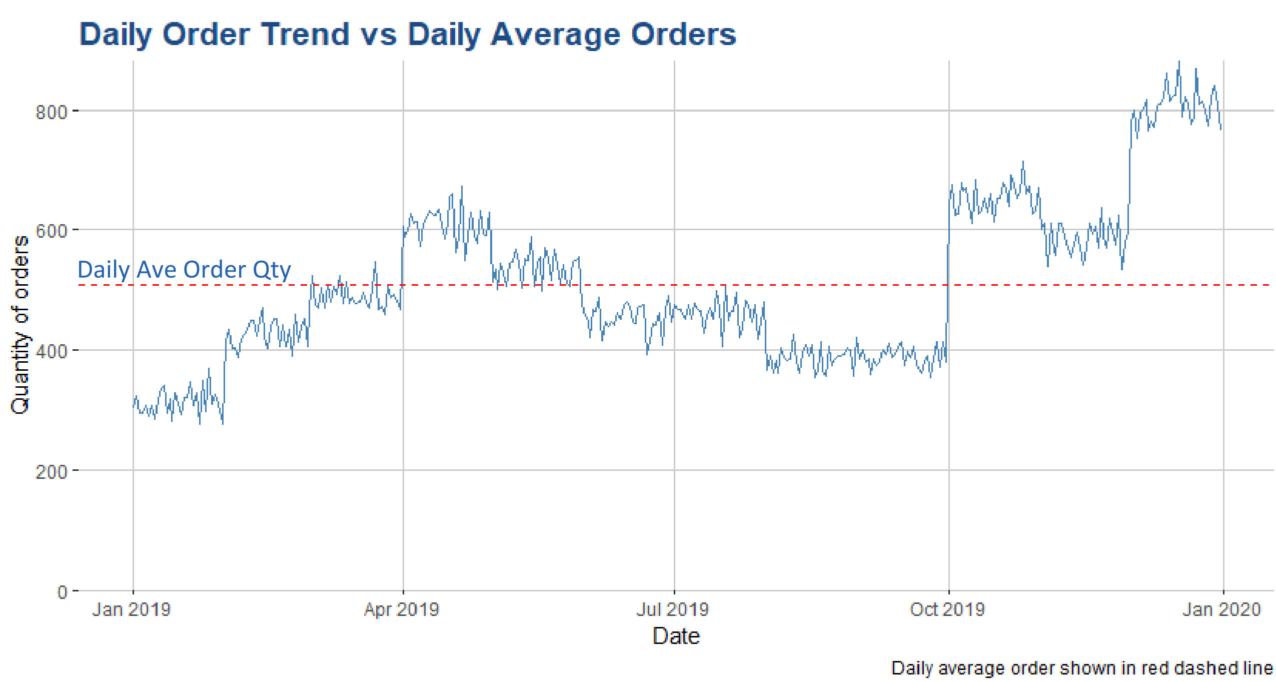
The monthly average sales in 2019 amounted to approximately 2.87 million USD. Notably, five months surpassed this average, including April, May, October, November, and December. The surge in total sales during the last quarter of the year aligns with the holiday season—a period of increased consumer spending.



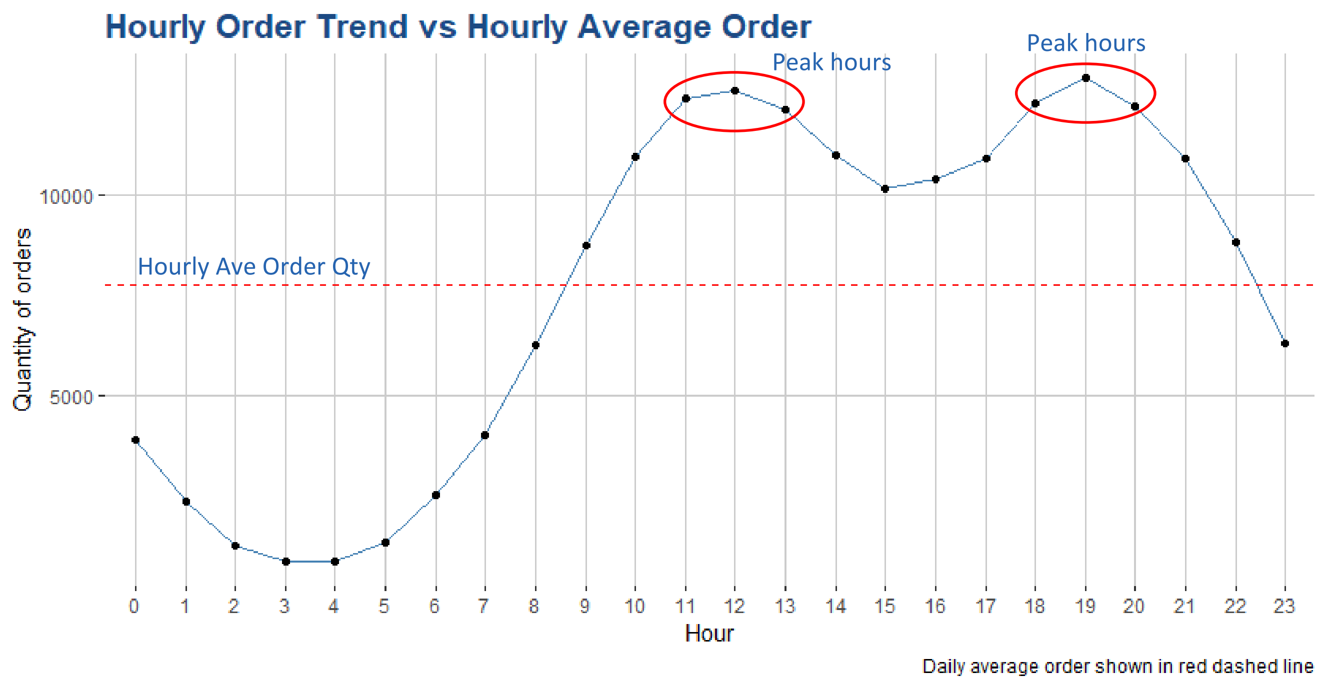
The trend of the monthly quantity of orders was also the same as the trend of monthly sales. On average, 17,428 orders were placed each month. December emerged as the top performer, with 28,137 orders, contributing a substantial 4.6 million USD in sales.



The daily order trend exhibited fluctuations, likely reflecting differences in shopping behavior between weekdays and weekends.



Examining the hourly order trend, we observed that the number of orders between 11 AM and 1 PM as well as 6 PM and 8 PM were remarkably high. These periods corresponded to lunchtime and dinnertime. Considering this, it would be prudent to run some digital marketing campaigns during these peak hours.

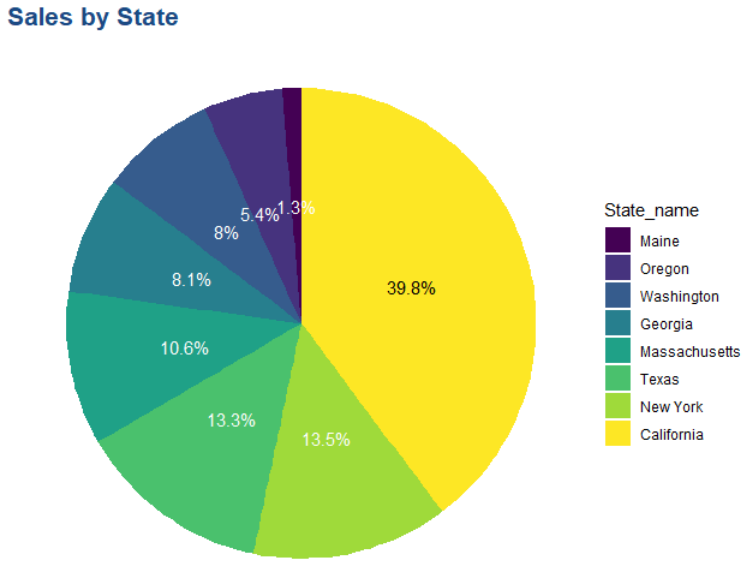


Geographic distribution

Dibs' customer base spanned 10 states across America. California led the way, accounting for 39.8% of total sales, followed by Texas and New York (both around 13.5%). Conversely, the states with low popularity like Maine and Oregon contributed the least at 1.3% and 5% respectively. In term of geographical efficiency, Dibs can locate its warehouse hubs near the high demand states, it can save the stock management and distribution cost.

Ideally, there should be 2 warehouses at least for the west and the east of US (in the red circle) and both can easily reach Texas.

The company can expand the business to some densely populated states which are close to current markets such as Florida, Pennsylvania, Illinois...

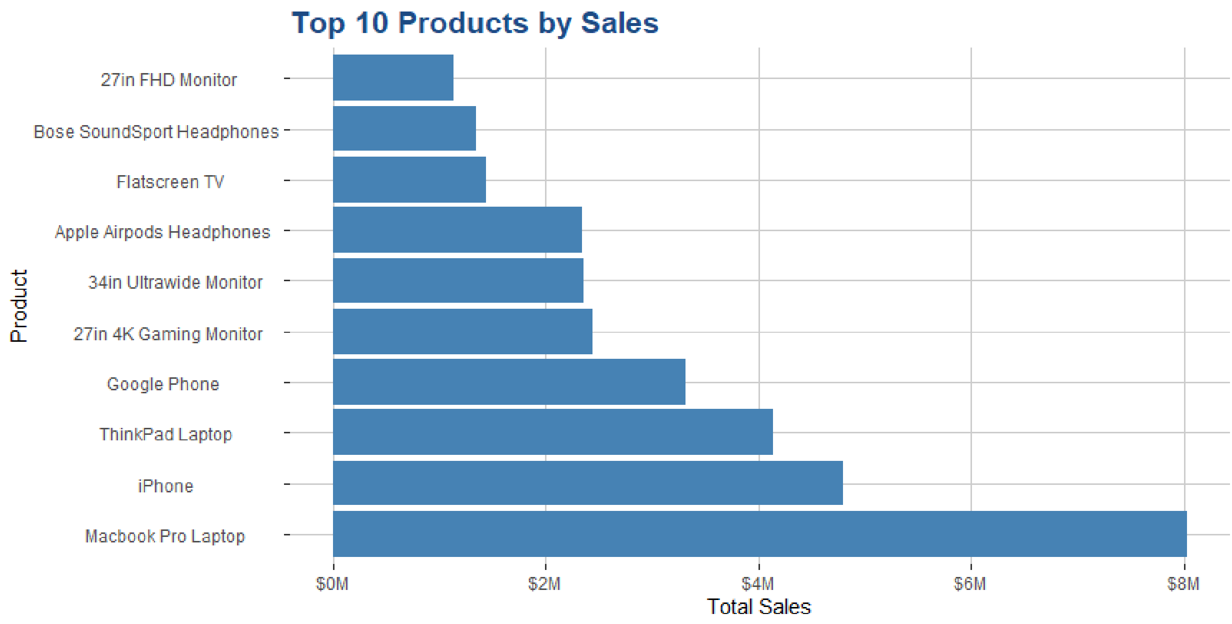


Product Offerings

Dibs has 19 SKU in the product portfolio including 12 main electronic devices and 7 accessories. As we can see from the table below, accessories are the products with high quantity of orders while electronic devices with higher price have lower quantity of orders.

Electronic devices			Accessories		
Product	Price (USD)	Qty Sold	Product	Price (USD)	Qty Sold
27in FHD Monitor	149.99	7,550	AAA Batteries (4-pack)	2.99	31,020
iPhone	700	6,850	AA Batteries (4-pack)	3.84	27,641
27in 4K Gaming Monitor	389.99	6,244	USB-C Charging Cable	11.95	23,977
34in Ultrawide Monitor	379.99	6,201	Lightning Charging Cable	14.95	23,223
Google Phone	600	5,533	Wired Headphones	11.99	20,571
Flatscreen TV	300	4,821	Apple Airpods Headphones	150	15,669
Macbook Pro Laptop	1700	4,728	Bose SoundSport Headphones	99.99	13,465
ThinkPad Laptop	999.99	4,132			
20in Monitor	109.99	4,129			
Vareebadd Phone	400	2,070			
LG Washing Machine	600	666			
LG Dryer	600	648			

Among 19 products, the MacBook Pro Laptop stood out, generating an impressive 8 million USD (representing 22% of total sales). Following were the iPhone, ThinkPad Laptop, and Google Phone—all premium-priced offerings.


Top 10 Products by Sales
27in FHD Monitor
Bose SoundSport Headphones
Flatscreen TV
Apple Airpods Headphones
34in Ultrawide Monitor
27in 4K Gaming Monitor
Google Phone
ThinkPad Laptop
iPhone
Macbook Pro Laptop
Product
$0M
$2M
$4M
$6M
$8M
Total Sales

PREDICTIVE MODELLING

On this section, we explore predictive modelling techniques to forecast sales quantities using historical data. We begin with an analysis of product clustering using K-means on quantity and unit price. The resultant clusters were then integrated into our forecast models which highlighted substantial linkages between unit quantity and price, aiding in enhancing our models.

Product Clustering with K-means

K-means clustering was used to group products according to comparable patterns in both quantity and price. This technique has also revealed significant linkages between quantity and price that were not immediately apparent. By leveraging on the cluster labels, it has allowed us to enhance both linear regression models and decision tree models. The analysis has also provided valuable insights into how inventory can be managed, and sales and marketing strategies can be tailored for each cluster.

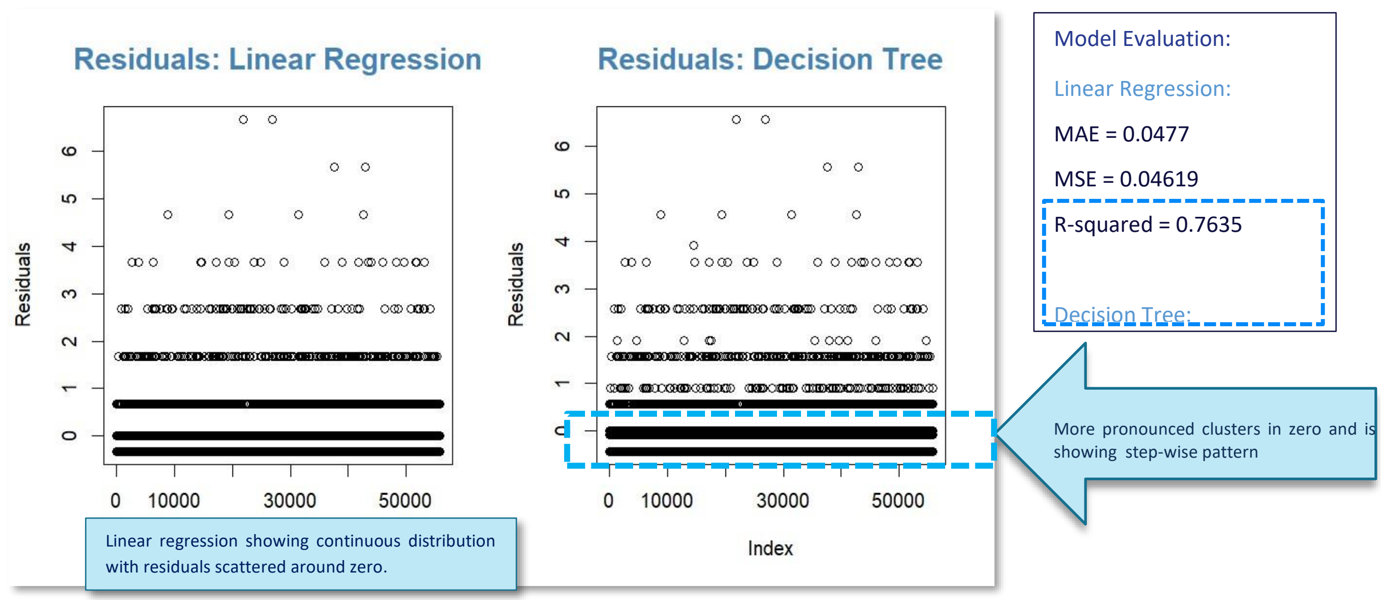
Cluster 1 High quantities at lower price range.

Cluster 2 Low quantities at higher price range.

Predictive Modelling Techniques

Two predictive modelling techniques were employed: Linear Regression and Decision Tree. Linear Regression, relying on a linear relationship between predictors (price and clusters) and the target variable (sales quantity). Linear regression offered interpretability, computational efficiency, however it is vulnerable to linear assumptions, outliers, and multicollinearity.

On the other hand, Decision Tree, a non-parametric model captures non-linear relationships, excelled in handling non-linear patterns and categorical variables but faced challenges with overfitting, instability, and bias-variance trade-off. These two models were tested and assessed which model will perform better. The residuals are calculated showing the difference between actual and predicted values and below are the results:



Quantitative Metrics: The Decision Tree model achieved the lowest MAE and MSE and higher Rsquared result, indicating it provides the most accurate predictions among the two models.

Qualitative Analysis: The residual analysis shows that the Decision Tree is handling the variability in the data and has a slight edge in error reduction.

Conclusion

The **Decision Tree model** is the most appropriate choice for predicting future sales quantity for the Dibs organization. It combines the benefits of improved accuracy from clustering and robust prediction performance as evidenced by the lowest MAE and MSE and better R-squared result. Moving forward, further tuning of the Decision Tree model and exploration of additional features and clustering strategies may enhance predictive accuracy even further.

INSIGHTS & RECOMMENDATIONS

Using the forecasted quantity from this model along with product price, we will be able to get the forecasted sales figures.

Based on our findings, we have formulated a set of targeted recommendations focused on inventory management, marketing, pricing, and sales strategies. These recommendations aim to optimize operations, enhance customer engagement, and drive revenue growth, ensuring Dibs remains competitive in the online retail market.

Inventory Management

Forecast demand: Using historical sales data to predict future demand, especially for peak seasons like 4th Quarter. Ensure high-demand items like AAA batteries are always well-stocked. We can use the predictive model that we have developed using Decision Tree to forecast sales quantities and periodically update the model to improve its accuracy.

Prioritization of Stock Levels: Focus on products at low price range with high quantities. These items should be stocked more heavily to avoid stockouts, especially if they have low to moderate prices suggesting high turnover.

Periodically review sales performance of high valued products with low sales quantity and consider reducing stock levels or promoting them differently if necessary. As an example, low selling item like LG Dryers can be reduced or discontinue to free up resources for higher demand products.

Sales Strategies

Seasonal Promotions: Plan and execute targeted promotions during peak sales months, particularly around December. Offer discounts, bundle deals, and special offers. High valued products with low sales quantity might benefit from promotional pricing strategies to increase their attractiveness and boost quantities sold.

Leverage the insights on best-selling product combinations (e.g., iPhone & Lightning Charging Cable) to create attractive bundle deals.

Time-Based Promotions: Peak hours are between 11AM to 1PM and 6PM to 8 PM. During these hours, we can run time-limited offers or flash sales to boost sales.

Marketing Strategies

Targeted Marketing: Focus marketing efforts on regions with the highest sales, such as San Francisco. Tailor campaigns to local preferences and purchasing behaviour.

Use data-driven insights to identify and target high-value customer segments effectively. Using K-Means we are able to see the cluster of high valued products with low quantities. We can deep dive on which customers are buying from these clusters and customized our programs accordingly.

Increased Visibility for Cluster 2: Products in this cluster may need increased marketing to boost awareness and demand. This could include email marketing campaigns, featured listings on websites, or in-store promotions.

Cross-Selling and Up-Selling: Implement strategies to promote products that are often sold together. For example, suggest Lightning Charging Cables with iPhone purchases. The electronic devices are usually paired with its accessories.

Use personalized recommendations based on past purchase behavior to increase average order value.

Advertising Strategies: Allocate a significant portion of the advertising budget during lunch and evening hours when sales peak. Use online advertising platforms to target ads during this time.

Process Improvement

Standardization of Product Names: Ensure consistency in product names by standardizing them across the dataset. This can involve using a predefined naming convention or mapping different variations of product names.

Opportunity for future expansion

Opportunity to expand the business to some densely populated states which are close to current markets such as Florida, Pennsylvania, Illinois. Proximity to an established sales hub will help the new site during its starting phase and will be able to easily share resources like warehouse and logistics as a business continuity plan in case the other site is not able to fulfill customer demand.